Bus 367 Homework 3

Due: November 12. The Homework should be sumbitted as a single pdf file on Brightspace.

1. Go to `https://www.nasdaq.com/market-activity/stocks` and download daily adjusted close prices for Apple (AAPL), Microsoft (MSFT) and Advanced Micro Devices (AMD) for the period from the first trading day of 2023 to the last trading day of September 2026. Compute the daily returns and plot the three time series on the same graph.
2. Compute the annualized mean return and standard deviation for each stock.
3. Compute the covariance and correlation matrix of returns.
4. Show the efficient frontier that can be obtained when only these three stocks are available.
5. Given the three assets (AAPL, MSFT and AMD), compute the weights that minimize portfolio variance for a given target return.
6. Show the efficient frontier when, besides the three stocks, you can borrow and lend at a risk-free rate of 2%.
7. For a target annual return of 7%, find the weights that minimize portfolio variance when you can use the three stocks and the risk-free rate.
8. Download, for the same time period, the daily returns on S&P 500 and Nasdaq. Suppose that the assets that you can use are Apple, Microsoft, AMD, a S&P 500 index and a Nasdaq index. Show the efficient frontier and compute the minimum variance portfolio if you want an expected return greater than 7%.
9. Repeat the exercise at the previous point assuming that you also have a risk-free asset paying 2%.
10. Construct a portfolio made up 50% by the Nasdaq index and by 50% by the S&P 500. Run a regression of the return on Apple on the portfolio return. Explain the results of the regression. Then do the same with Microsoft and for AMD.